

Question 47:

a) Create EMP table with ECODE(primary key), ENAME, DCODE, GRADE, BASIC & JN-DT as the columns.[except BASIC & JN-DT,all columns are of char type and size of Grade is 1.]

Query :

```
create table emp( ecode char(4) primary key, ename varchar(40), dcode char(4),  
grade char(1), basic decimal(9,2), jn_dt date);
```

Output :

```
mysql> create table emp(  
-> ecode char(4) primary key,  
-> ename varchar(40),  
-> dcode char(4),  
-> grade char(1),  
-> basic decimal(9,2),  
-> jn_dt date  
-> );  
Query OK, 0 rows affected (0.03 sec)
```

b)Insert number of rows.

Query :

```
INSERT INTO EMP (ECODE, ENAME, DCODE, GRADE, BASIC, JN_DT) VALUES
```

```
('E001','RAHUL','D001','A',50000.00,'2022-01-15'),  
('E002','AMIT','D002','B',45000.00,'2021-06-20'),  
('E003','PRIYA','D001','A',55000.00,'2020-03-10'),  
('E004','SNEHA','D003','C',35000.00,'2023-02-05'),  
('E005','ROHAN','D002','B',48000.00,'2021-11-18'),  
('E006','ANITA','D004','A',62000.00,'2019-08-12'),  
('E007','VIKAS','D003','B',42000.00,'2020-12-01'),  
('E008','POOJA','D001','A',58000.00,'2022-07-25'),  
('E009','ARJUN','D005','C',33000.00,'2023-04-11'),  
('E010','NEHA','D002','B',47000.00,'2021-09-14'),  
('E011','KARAN','D004','A',61000.00,'2018-11-30'),  
('E012','RITU','D005','C',36000.00,'2024-01-20'),  
('E013','MOHAN','D003','B',44000.00,'2020-05-08'),  
('E014','NISHA','D001','A',59000.00,'2022-03-19'),  
('E015','DEEPAK','D002','B',49000.00,'2021-10-27'),  
('E016','SONALI','D004','A',65000.00,'2019-02-16'),  
('E017','MANOJ','D005','C',34000.00,'2023-08-09'),  
('E018','KAVITA','D003','B',43000.00,'2020-06-22'),  
('E019','AJAY','D001','A',57000.00,'2022-12-05'),  
('E020','MEENA','D002','B',46000.00,'2021-04-17');
```

Output :

```
mysql> INSERT INTO EMP (ECODE, ENAME, DCODE, GRADE, BASIC, JN_DT) VALUES  
-> ('E001','RAHUL','D001','A',50000.00,'2022-01-15'),  
-> ('E002','AMIT','D002','B',45000.00,'2021-06-20'),  
-> ('E003','PRIYA','D001','A',55000.00,'2020-03-10'),  
-> ('E004','SNEHA','D003','C',35000.00,'2023-02-05'),  
-> ('E005','ROHAN','D002','B',48000.00,'2021-11-18'),  
-> ('E006','ANITA','D004','A',62000.00,'2019-08-12'),  
-> ('E007','VIKAS','D003','B',42000.00,'2020-12-01'),  
-> ('E008','POOJA','D001','A',58000.00,'2022-07-25'),  
-> ('E009','ARJUN','D005','C',33000.00,'2023-04-11'),  
-> ('E010','NEHA','D002','B',47000.00,'2021-09-14'),  
-> ('E011','KARAN','D004','A',61000.00,'2018-11-30'),  
-> ('E012','RITU','D005','C',36000.00,'2024-01-20'),  
-> ('E013','MOHAN','D003','B',44000.00,'2020-05-08'),  
-> ('E014','NISHA','D001','A',59000.00,'2022-03-19'),  
-> ('E015','DEEPAK','D002','B',49000.00,'2021-10-27'),  
-> ('E016','SONALI','D004','A',65000.00,'2019-02-16'),  
-> ('E017','MANOJ','D005','C',34000.00,'2023-08-09'),  
-> ('E018','KAVITA','D003','B',43000.00,'2020-06-22'),  
-> ('E019','AJAY','D001','A',57000.00,'2022-12-05'),  
-> ('E020','MEENA','D002','B',46000.00,'2021-04-17');  
Query OK, 20 rows affected (0.01 sec)  
Records: 20 Duplicates: 0 Warnings: 0
```

Question 48 : change the column heading as shown below, So that in subsequent SELECT statement newly set heading will be shown:

ECODE EMPLOYEE CODE

ENAME NAME

DCODE DEPT.CODE

JN-DT JOINING DATE

Query :

```
SELECT
    ECODE AS 'EMPLOYEE CODE',
    ENAME AS 'NAME',
    DCODE AS 'DEPT.CODE',
    JN_DT AS 'JOINING DATE'
FROM EMP;
```

Output :

```
mysql> SELECT
->     ECODE AS 'EMPLOYEE CODE',
->     ENAME AS 'NAME',
->     DCODE AS 'DEPT.CODE',
->     JN_DT AS 'JOINING DATE'
-> FROM EMP;
```

EMPLOYEE CODE	NAME	DEPT.CODE	JOINING DATE
E001	RAHUL	D001	2022-01-15
E002	AMIT	D002	2021-06-20
E003	PRIYA	D001	2020-03-10
E004	SNEHA	D003	2023-02-05
E005	ROHAN	D002	2021-11-18
E006	ANITA	D004	2019-08-12
E007	VIKAS	D003	2020-12-01
E008	POOJA	D001	2022-07-25
E009	ARJUN	D005	2023-04-11
E010	NEHA	D002	2021-09-14
E011	KARAN	D004	2018-11-30
E012	RITU	D005	2024-01-20
E013	MOHAN	D003	2020-05-08
E014	NISHA	D001	2022-03-19
E015	DEEPAK	D002	2021-10-27
E016	SONALI	D004	2019-02-16
E017	MANOJ	D005	2023-08-09
E018	KAVITA	D003	2020-06-22
E019	AJAY	D001	2022-12-05
E020	MEENA	D002	2021-04-17

20 rows in set (0.00 sec)

Question 49: Set the format of columns as mentioned below, So that in subsequent SELECT statement, values appear in the specified format:

***format of BASIC is such that a value of 7000 will be shown as 7,000**

***Format of GRADE will be such that full column name appears in the display.**

***For JN-DT format is such that 01-JAN-00 will be shown as JANURY 01,2000**

Query :

```
SELECT
    ECODE AS 'EMPLOYEE CODE',
    ENAME AS 'NAME',
    DCODE AS 'DEPT.CODE',
    GRADE AS 'GRADE',
    FORMAT(BASIC,0) AS 'BASIC',
    DATE_FORMAT(JN_DT,'%M %d,%Y') AS 'JOINING DATE'
FROM EMP;
```

Output :

```
mysql> SELECT
->     ECODE AS 'EMPLOYEE CODE',
->     ENAME AS 'NAME',
->     DCODE AS 'DEPT.CODE',
->     GRADE AS 'GRADE',
->     FORMAT(BASIC,0) AS 'BASIC',
->     DATE_FORMAT(JN_DT,'%M %d,%Y') AS 'JOINING DATE'
-> FROM EMP;
```

EMPLOYEE CODE	NAME	DEPT.CODE	GRADE	BASIC	JOINING DATE
E001	RAHUL	D001	A	50,000	January 15,2022
E002	AMIT	D002	B	45,000	June 20,2021
E003	PRIYA	D001	A	55,000	March 10,2020
E004	SNEHA	D003	C	35,000	February 05,2023
E005	ROHAN	D002	B	48,000	November 18,2021
E006	ANITA	D004	A	62,000	August 12,2019
E007	VIKAS	D003	B	42,000	December 01,2020
E008	POOJA	D001	A	58,000	July 25,2022
E009	ARJUN	D005	C	33,000	April 11,2023
E010	NEHA	D002	B	47,000	September 14,2021
E011	KARAN	D004	A	61,000	November 30,2018
E012	RITU	D005	C	36,000	January 20,2024
E013	MOHAN	D003	B	44,000	May 08,2020
E014	NISHA	D001	A	59,000	March 19,2022
E015	DEEPAK	D002	B	49,000	October 27,2021
E016	SONALI	D004	A	65,000	February 16,2019
E017	MANOJ	D005	C	34,000	August 09,2023
E018	KAVITA	D003	B	43,000	June 22,2020
E019	AJAY	D001	A	57,000	December 05,2022
E020	MEENA	D002	B	46,000	April 17,2021

20 rows in set (0.00 sec)

Question 50:

- Show the display attributes of all the columns.
- Show the display attributes of particular column.
- Suppress the newly set attributes of JN-DT .Try a select statement.
- Reset the newly set attributes of JN-DT
- Reset the newly set attributes of all columns.
- Shown the display attributes of all columns.

Query :

DESCRIBE EMP;

Output :

mysql> DESCRIBE EMP;

Field	Type	Null	Key	Default	Extra
ecode	char(4)	NO	PRI	NULL	
ename	varchar(40)	YES		NULL	
dcode	char(4)	YES		NULL	
grade	char(1)	YES		NULL	
basic	decimal(9,2)	YES		NULL	
jn_dt	date	YES		NULL	

6 rows in set (0.01 sec)

Question 51:

- Show the records from EMP table in the ascending order of DCODE . DCODE value will be shown only for the first record of that department (same of DCODE is not repeated)

Query :

SELECT * FROM EMP ORDER BY DCODE;

Output :

mysql> SELECT * FROM EMP ORDER BY DCODE;

ecode	ename	dcode	grade	basic	jn_dt
E001	RAHUL	D001	A	50000.00	2022-01-15
E003	PRIYA	D001	A	55000.00	2020-03-10
E008	POOJA	D001	A	58000.00	2022-07-25
E014	NISHA	D001	A	59000.00	2022-03-19
E019	AJAY	D001	A	57000.00	2022-12-05
E002	AMIT	D002	B	45000.00	2021-06-20
E005	ROHAN	D002	B	48000.00	2021-11-18
E010	NEHA	D002	B	47000.00	2021-09-14
E015	DEEPAK	D002	B	49000.00	2021-10-27
E020	MEENA	D002	B	46000.00	2021-04-17
E004	SNEHA	D003	C	35000.00	2023-02-05
E007	VIKAS	D003	B	42000.00	2020-12-01
E013	MOHAN	D003	B	44000.00	2020-05-08
E018	KAVITA	D003	B	43000.00	2020-06-22
E006	ANITA	D004	A	62000.00	2019-08-12
E011	KARAN	D004	A	61000.00	2018-11-30
E016	SONALI	D004	A	65000.00	2019-02-16
E009	ARJUN	D005	C	33000.00	2023-04-11
E012	RITU	D005	C	36000.00	2024-01-20
E017	MANOJ	D005	C	34000.00	2023-08-09

20 rows in set (0.00 sec)

- b) Further take measures so that ,after displaying the records of a department it skips one line .
 c)Further take measures so that records one also ordered on the basic of GRADE with in a department & same GRADE value is not repeated.

Query :

```
SELECT * FROM EMP ORDER BY DCODE, GRADE;
```

Output :

```
mysql> SELECT * FROM EMP ORDER BY DCODE, GRADE;
```

ecode	ename	dcode	grade	basic	jn_dt
E001	RAHUL	D001	A	50000.00	2022-01-15
E003	PRIYA	D001	A	55000.00	2020-03-10
E008	POOJA	D001	A	58000.00	2022-07-25
E014	NISHA	D001	A	59000.00	2022-03-19
E019	AJAY	D001	A	57000.00	2022-12-05
E002	AMIT	D002	B	45000.00	2021-06-20
E005	ROHAN	D002	B	48000.00	2021-11-18
E010	NEHA	D002	B	47000.00	2021-09-14
E015	DEEPAK	D002	B	49000.00	2021-10-27
E020	MEENA	D002	B	46000.00	2021-04-17
E007	VIKAS	D003	B	42000.00	2020-12-01
E013	MOHAN	D003	B	44000.00	2020-05-08
E018	KAVITA	D003	B	43000.00	2020-06-22
E004	SNEHA	D003	C	35000.00	2023-02-05
E006	ANITA	D004	A	62000.00	2019-08-12
E011	KARAN	D004	A	61000.00	2018-11-30
E016	SONALI	D004	A	65000.00	2019-02-16
E009	ARJUN	D005	C	33000.00	2023-04-11
E012	RITU	D005	C	36000.00	2024-01-20
E017	MANOJ	D005	C	34000.00	2023-08-09

20 rows in set (0.00 sec)

- d) Take measure so that at the end of each GRADE in a department it will show average Basic for that grade in that department. At the end of each department, it will show the average & total Basic for the department. At the end of all departments it will show the overall total basic & average basic.

Query :

```
SELECT
    DCODE,
    GRADE,
    AVG(BASIC) AS AVG_BASIC
FROM EMP
GROUP BY DCODE, GRADE
ORDER BY DCODE, GRADE;
```

Output :

```
mysql> SELECT
->     DCODE,
->     GRADE,
->     AVG(BASIC) AS AVG_BASIC
-> FROM EMP
-> GROUP BY DCODE, GRADE
-> ORDER BY DCODE, GRADE;
```

DCODE	GRADE	AVG_BASIC
D001	A	55800.000000
D002	B	47000.000000
D003	B	43000.000000
D003	C	35000.000000
D004	A	62666.666667
D005	C	34333.333333

6 rows in set (0.01 sec)